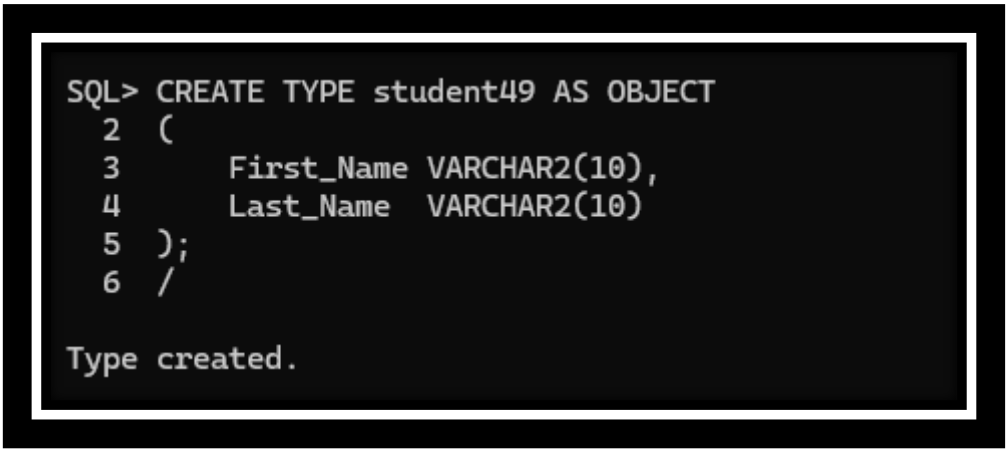
Assignment No	**3**
Title	**ORDBMS**
Objective	**Creating Abstract Data Types and Using Them in Tables**
Roll No	**MCA2649**

Source Code:-

```
CREATE TYPE student46 AS OBJECT
(
   First_Name VARCHAR2(10),
   Last_Name  VARCHAR2(10)
);
/
```

Output:-



```
SQL> CREATE TABLE student_detail
  2  (
  3       PName student49
  4  );

Table created.
```

```
SQL> INSERT INTO student_detail
  2  VALUES (student49('Satish', 'Rathod'));

1 row created.

SQL> INSERT INTO student_detail
  2  VALUES (student49('Niranjan', 'Amrute'));

1 row created.
```

```
SQL> SELECT P.PName.First_Name
  2  FROM student_detail P;

PNAME.FIRS
----------
Satish
Niranjan
Khushi
Saanvi
Aarya
```

```
SQL> SELECT P.PName.First_Name || ' ' || P.PName.Last_Name
  2  FROM student_detail P;

P.PNAME.FIRST_NAME||'
--------------------
Satish Rathod
Niranjan Amrute
Khushi Pujari
Saanvi Sawant
Aarya Jadhav
```

```
SQL> SELECT P.PName.First_Name || '@@@@@' || P.PName.Last_Name
  2  FROM student_detail P;

P.PNAME.FIRST_NAME||'@@@@
------------------------
Satish@@@@@Rathod
Niranjan@@@@@Amrute
Khushi@@@@@Pujari
Saanvi@@@@@Sawant
Aarya@@@@@Jadhav
```

```
SQL> CREATE TYPE Investment_Type49 AS OBJECT
  2  (
  3       Year NUMBER,
  4       MEMBER FUNCTION Prod(Invest NUMBER)
  5       RETURN NUMBER
  6  );
  7  /

Type created.
```

```
SQL> CREATE TYPE BODY Investment_Type49 AS
  2       MEMBER FUNCTION Prod(Invest NUMBER)
  3       RETURN NUMBER
  4       IS
  5       BEGIN
  6           RETURN (Year + Invest);
  7       END;
  8  END;
  9  /

Type body created.
```

```
SQL> CREATE TABLE Investment_Details
  2  (
  3       ID NUMBER,
  4       Details Investment_Type49
  5  );

Table created.
```

```
SQL> INSERT INTO Investment_Details
  2  VALUES (1, Investment_Type49(2021));

1 row created.

SQL>
SQL> INSERT INTO Investment_Details
  2  VALUES (2, Investment_Type49(2022));

1 row created.
```

```
SQL> COMMIT;

Commit complete.
```

```
SQL>
SQL> CREATE OR REPLACE TYPE BODY Investment_Type49 AS
  2       MEMBER FUNCTION Prod(Invest NUMBER)
  3       RETURN NUMBER
  4       IS
  5       BEGIN
  6           RETURN (Year + Invest);
  7       END;
  8  END;
  9  /

Type body created.
```

```
SQL> CREATE TYPE Name49 AS OBJECT
  2  (
  3       First_Name VARCHAR2(10),
  4       Last_Name  VARCHAR2(10)
  5  );
  6  /

Type created.
```

```
SQL> CREATE TYPE Address49 AS OBJECT
  2  (
  3       Street VARCHAR2(10),
  4       City   VARCHAR2(10)
  5  );
  6  /

Type created.
```

```
SQL> CREATE TABLE People49
  2  (
  3       Name  Name49,
  4       Addrs Address49,
  5       DOB   DATE
  6  );

Table created.
```

```
SQL> INSERT INTO People49
  2  VALUES (
  3       Name49('Satish', 'Rathod'),
  4       Address49('Mira Road', 'Thane'),
  5       TO_DATE('02-06-1997', 'DD-MM-YYYY')
  6  );

1 row created.
```

```
SQL> select * from People49
  2  ;

NAME(FIRST_NAME, LAST_NAME)
--------------------------------------------------------------------------------
ADDRS(STREET, CITY)
--------------------------------------------------------------------------------
DOB
---------
NAME49('Satish', 'Rathod')
ADDRESS49('Mira Road', 'Thane')
02-JUN-97
```

```
SQL> SELECT pd.Name.First_Name || ' ' || pd.Name.Last_Name
  2  FROM People49 pd;

PD.NAME.FIRST_NAME||'
--------------------
Satish Rathod
```

```
SQL> SELECT pd.Addrs.Street || ' ' || pd.Addrs.City
  2  FROM People49 pd;

PD.ADDRS.STREET||''||
--------------------
Mira Road Thane
```

```
SQL> CREATE TYPE Demo AS OBJECT
  2  (
  3       Id NUMBER,
  4       MEMBER FUNCTION get_square RETURN NUMBER
  5  );
  6  /

Type created.
```

```
SQL> CREATE TABLE Demo_Tbl
  2  (
  3       Col Demo
  4  );

Table created.
```

```
SQL> INSERT INTO Demo_Tbl
  2  VALUES (Demo(5));

1 row created.
```

```
SQL> create type body Demo
  2  is member function get_square
  3  return number
  4  is n number;
  5  begin
  6  select s.Col.ID*s.Col.ID into n
  7  From Demo_Tbl s;
  8  return(n);
  9  end;
 10  end;
 11  /
```

```
SQL> select v.Col.get_square() from Demo_Tbl v;

V.COL.GET_SQUARE()
------------------
                 9
                25
```